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Total Solar Eclipse: Studying the Sun's 'Invisible' Part

Total solar eclipse: The best time to study Sun's 'invisible' part

Anjali Marar

Bengaluru, August 12

EUROPE WITNESSED its first total solar eclipse in 27 years between Wednesday night and Thursday morning (India time).

A solar eclipse occurs when the Moon passes between the Sun and the Earth, blocking out some or all of the star's light. The latter phenomenon happens during a total solar eclipse. It is a rare occurrence, requiring the Sun, the Moon and the Earth to line-up perfectly in space.

While not all of Europe will experience "totality" — the phase when the Moon completely covers the Sun's surface — large parts of the continent will still experience near night-like conditions.

A solar eclipse also happens to be the best time for scientists to study the Sun. Here's what they study at that time — and how they do it.

Why do we need to study the Sun?

For scientists, observing the Sun's corona and the inner corona is crucial. The corona's magnetic field triggers solar flares and "coronal mass ejections", which can

disrupt satellites, GPS and other communication systems on Earth.

The corona is the Sun's outermost atmosphere, extending millions of kilometres into space. The inner corona is the part of the Sun's outer atmosphere that is closest to the star's surface.

However, it is extremely faint. To study it, scientists need to block the solar disc artificially using an instrument called a coronagraph, which is usually placed on solar probes or missions. On many occasions, however, these instruments are affected by scattered light — making it impossible to observe the inner corona.

How does a solar eclipse help?

During a total solar eclipse, the Moon completely blocks the solar disc, making the inner corona visible in the form of a halo around the lunar body.

This provides helio-physicists with a valuable window to make ground-based observations of the solar atmosphere.

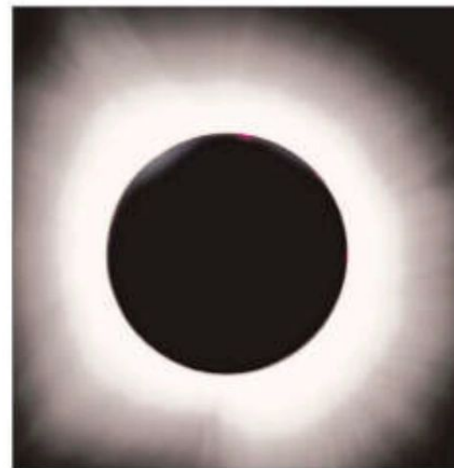
"When an eclipse happens, the bright inner disk gets blocked and the fainter outer atmosphere gets revealed, which is the solar corona. The eclipse thus gives us

Path of totality

- The total eclipse will be visible along the north of Russia, Greenland, west Iceland, northern Spain and a small area in northeast Portugal.

- Together, these locations form the 'path of totality' — the narrow strip on the Earth's surface where the Moon can be seen completely blocking the Sun.

- In any total solar eclipse, the duration of 'totality' — the phase when the Moon completely covers the Sun's surface — is the longest near the centre of the path of totality.



A total solar eclipse seen in Texas in 2024. The Sun's corona can be seen as a halo. AP

clues to the underlying magnetic structure of the Sun's corona," Dibyendu Nandi, an astrophysicist from the Raman Research Institute, told *The Indian Express*.

Another factor to consider is the solar cycle — the roughly 11-year cycle during which the Sun's activity rises and falls, marked by changes in the number of sun-

spots, solar flares and other activity.

In the current solar cycle, which began in 2020, the Sun has already passed its maxima phase — the period when it is most active and when more structures in the inner corona are likely to be visible.

Owing to these reasons, scientists plan multiple experiments and solar observations during total solar eclipses, particularly from the location where the maximum eclipse is expected.

What can scientists observe?

During the eclipse, an Indian group of researchers at Centre of Excellence for Space Science India (CESSI), IISER-Kolkata, has predicted the possibility of spotting multiple large-scale streamers, which are petal-like structures. "These are nothing but bright magnetic field structures rooted to the solar surface. There is no other way to see the magnetic fields, because the Sun's disc is bright. During a natural eclipse, it is possible to see the Sun's corona from the ground, too," Nandi said.

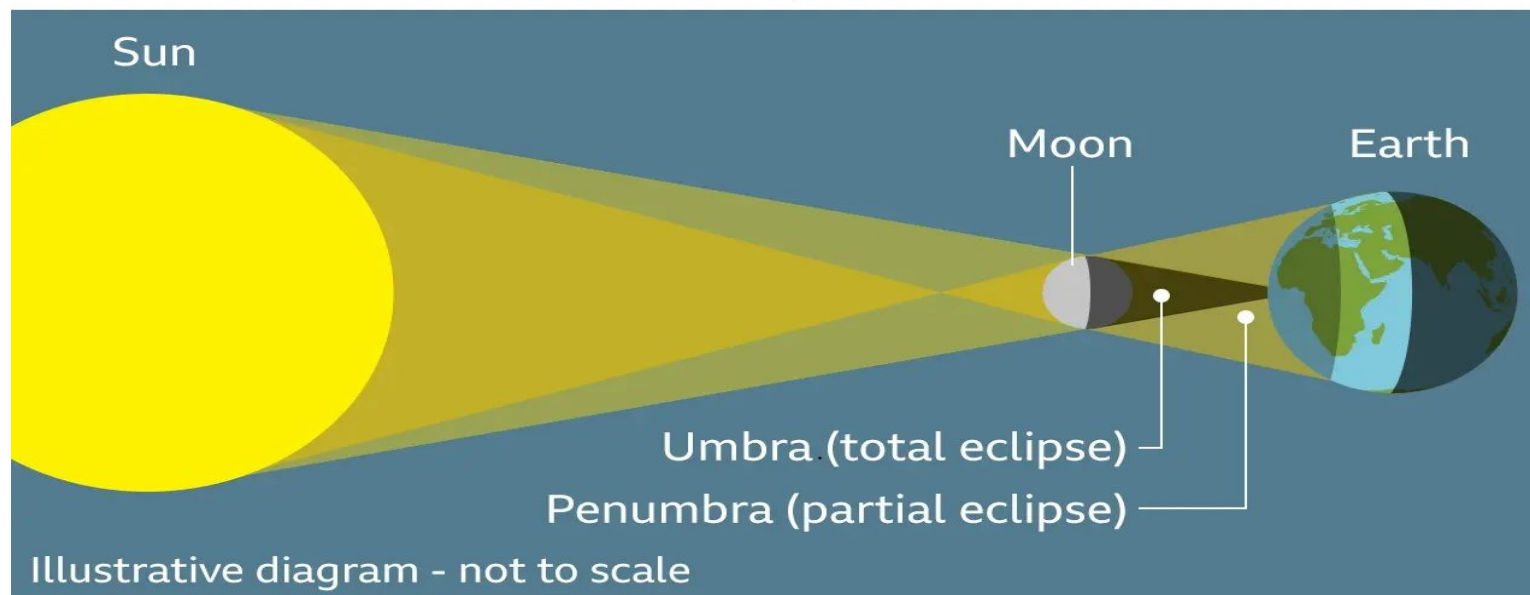
If these magnetic field structures are spotted during the event, scientists expect them to be highly complex and spectacular.

Total Solar Eclipse: Studying the Sun's 'Invisible' Part

- **Solar Eclipse**

- **Alignment:** A total solar eclipse occurs when the Sun, Moon and Earth are aligned in a near-straight line, a configuration called syzygy.
- **The Moon's shadow has two parts:** the umbra, where sunlight is completely blocked and totality occurs, and the penumbra, where sunlight is only partially blocked, producing a partial eclipse.

Shadows of a solar eclipse



Supreme Court Clarifies Scope of Police Custody Under BNSS

Supreme Court clarifies scope of police custody under BNSS

Court held that Section 187(2) of BNSS enlarges the window during which police custody may be sought

R.K. Vij

The story so far:

The Supreme Court recently held in *The State of Andhra Pradesh vs Suda Suresh Veera Venkata Naga Raju* that Section 187(2) of the Bharatiya Nagarik Suraksha Sanhita (BNSS), enlarges the window during which police custody may be sought by the investigating agency. Such custody, under the BNSS is now available in parts, though in aggregate not exceeding 15 days during the first 40 or 60 days of the total permissible period of detention, rather than being confined to the first 15 days of remand alone.

Case under consideration by the SC

In a case of custodial death in which the dead body of the deceased-victim remained untraced, the original hard disks of the CCTV system were yet to be

discovered, and discoveries under the Bharatiya Sakshya Adhiniyam (BSA) were imminent, the judicial magistrate granted eight days of police custody, and allowed the accused to nominate two advocates of his choice to be present during custody. Out of the two advocates, one was allowed to be present at any given point of time during investigation. The order stated that 'under no circumstances shall the period of police custody granted under this order be extended'.

While on appeal, the Andhra Pradesh High Court modified some conditions pertaining to the period of police custody and presence of an advocate throughout the probe. The State therefore challenged the above conditions in the Supreme Court on the premise that these directly impinged upon the rights of the investigating officer to conduct unhindered investigation.

The Supreme Court held that the

apprehensions expressed by the investigating agency were justified and the statutory scheme under the BNSS which enlarges the window of police custody, must also be borne in mind. A magistrate or a court cannot place an absolute and non-extendable outer limit on custody, since such a limit forecloses recourse to Section 187(2) of the BNSS.

The Court therefore permitted 7 days of (additional) police custody so that the total period of police remand would not exceed 15 days.

Current law on police remand

Section 58 of the BNSS provides that any person arrested by the police without warrant cannot be detained for more than 24 hours, unless authorised by a magistrate under Section 187.

Section 187(2) of the BNSS provides that in case it appears to the investigating officer that the probe cannot be

completed within 24 hours, and there are grounds for believing that the accusation is well founded, the magistrate may authorise the detention of the accused for a period not exceeding 15 days in the whole, or in parts, at any time during the initial 40 days or 60 days out of the detention period of 60 days or 90 days.

Section 187(3) provides that the magistrate may authorise the (judicial) detention of an accused person beyond the period of 15 days, if he is satisfied that adequate grounds exist for doing so, for a period exceeding - (i) 90 days, where the investigation relates to an offence punishable with death, imprisonment for life or imprisonment for a term of 10 years or more; (ii) 60 days, where the investigation relates to any other offence. Thus, in case the probe is not completed within the above stipulated time period, the accused person may be released, which is widely known as default bail.

The Supreme Court in the *Suda Suresh* case held that this legislative change was intended to meet situations where fresh facts, discoveries or leads may emerge during the course of investigation.

Therefore, the window of police remand was enlarged beyond the initial 15 days, up to 40 days or 60 days of investigation.

Previously, Section 167 of the CrPC provided that the detention of an accused in police custody could not be granted beyond the initial 15 days in the whole.

However, during this first period of 15 days the magistrate could alter the nature of the custody from judicial custody to police custody and vice versa.

On the presence of an advocate

Section 38 of the BNSS provides that 'when any person is arrested and interrogated by the police, he shall be entitled to meet an advocate of his choice during interrogation, though not throughout interrogation'.

The top court held that a plain reading of the provision makes it clear that it does not contemplate the continuous, ongoing physical presence of an advocate for the entirety of each interrogation session. The Court held that an unqualified entitlement to continuous presence, would travel beyond what Section 38 BNSS itself contemplates. The Court therefore modified the condition imposed by the High Court, and held that such a lawyer shall only be allowed to remain present within the site of interrogation where he can see the respondent-accused.

The Court also held that instead of uninterrupted videography of the transit of the accused between locations, the audio-visual recording of the actual interrogation, and of any proceedings of discovery or recovery undertaken with the accused, will satisfy the requirement. (R.K. Vij is a former IPS officer.)

Supreme Court Clarifies Scope of Police Custody Under BNSS

Police Custody refers to the detention of an accused person in a police station for the purpose of investigation. In this type of custody, the accused remains under the direct control of the police authorities. Police custody is generally granted by a Magistrate for a limited period to allow the police to interrogate the accused and collect evidence.

- The accused is kept in a police station lock-up.
- The police have direct physical control over the accused.
- Main purpose is interrogation and investigation.
- Maximum duration is 15 days from the date of first remand as per Section 167 of the Criminal Procedure Code.
- The accused must be produced before a Magistrate within 24 hours of arrest.

Judicial Custody refers to the detention of an accused person in a jail under the supervision of a Magistrate. In this type of custody, the accused is not under direct police control but is kept in a prison. Judicial custody is usually ordered when the Magistrate believes that further detention is necessary but does not require police interrogation.

- The accused is lodged in a jail or prison.
- The jail authorities are responsible for the custody of the accused.
- Police cannot interrogate the accused without court permission.
- Judicial custody can extend up to 90 days for serious offences and 60 days for less serious offences.
- It ensures greater protection of the rights of the accused.

Supreme Court Clarifies Scope of Police Custody Under BNSS

Regarding India, consider the following statements :

1. Judicial custody means an accused is in the custody of the concerned magistrate and such accused is locked up in a police station, not in jail.
 2. During judicial custody, the police officer in charge of the case is not allowed to interrogate the suspect without the approval of the court.
- which of the statements given above is/are correct?

This question was previously asked in

UPSC Civil Services Exam
(Prelims) General Studies Offici...

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1. 1 only

2. 2 only

3. Both 1 and 2

4. Neither 1 nor 2

A Timely Reset for the Food Security Act

A timely reset for the Food Security Act

India's food security debate has entered a new phase. Estimates based on India's latest household consumption survey suggest that the share of households unable to afford the ICMR-National Institute of Nutrition (NIN) recommended diet fell from about 52% in 2011-12 to about 25% in 2023-24, affecting 25% of rural and 21% of urban households. This is substantial progress, but crores of households still cannot afford an adequate healthy diet. Poverty, hunger and inability to afford a healthy diet overlap; a modern food-security policy must recognise that they are not identical.

The draft National Food Security (Amendment) Bill, 2026 links Antyodaya Anna Yojana (AAY) entitlements to household size. The policy challenge is to correct inequity without reducing existing foodgrain access, while creating a pathway towards nutrition security.

Designing a fairer entitlement

Priority Households receive 5 kg of foodgrains per person per month, while AAY households receive a flat 35 kg per household. The latter protects smaller and highly vulnerable families but provides progressively less per person as household size grows: a seven-member household receives 5 kg per person and an eight-member household about 4.4 kg, below the Priority Household entitlement. The draft proposes 7 kg per person, capped at 35 kg. This would reduce support for households with one to four members by 20% to 80%, leave those with five or more unchanged, and provide no AAY household with additional foodgrain.

Tamil Nadu illustrates the potential impact. The State government reports that 15.75 lakh of its 18.64 lakh AAY households (84.5%) have fewer than five members. It estimates that the proposal would reduce the monthly AAY allocation from 65,261 tonnes to 42,040 tonnes, a fall of about 35.6%. Although specific to Tamil Nadu, these projections show why household composition and vulnerability must be considered: a smaller household may include an older person living alone, a widow or a person with disability.

As the proposal is taken forward, the existing 35 kg entitlement should be preserved through an explicit no-loss safeguard. Because retaining 35 kg as both the minimum and maximum would leave the per-person formula with no practical effect, support beyond 35 kg for larger or highly dependent households should be examined as a separate policy question using household-consumption evidence, nutritional requirements and transparent fiscal costing.

Entitlement reform must also address coverage. Although the NFSA allows coverage of up to 75% of the rural and 50% of the urban population, its 81.35-crore beneficiary ceiling remains based on Census 2011: about 80 crore people receive free foodgrains. Against the estimated population of 146.4 crore in 2025, this ceiling covers only 55.6%. It should be recalculated when the Census 2027 figures become available. Meanwhile, updated

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The proposed amendment offers an opportunity to preserve the 35 kg AAY entitlement, assess the needs of larger households, and finance a gradual transition toward more diverse and nutritious diets

population estimates and deprivation criteria could support a transparent review, with accessible inclusion and appeal mechanisms.

Protecting grain while diversifying diets

Protecting the grain entitlement does not mean preserving a cereal-only understanding of food security. NFHS-6 (2023-24) found that stunting among children under five fell from 35.5% to 29.3%, but wasting and underweight barely changed, from 19.3% to 19.0% and 32.1% to 31.8%, respectively. Only about 15% of children (six-23 months) receive a minimally adequate diet. Meanwhile, the ICMR-India Diabetes (INDIAB) study estimated that 101 million Indians had diabetes and 136 million had prediabetes in 2021. These figures reflect India's double burden of malnutrition. Maternal undernutrition and low birth weight are associated with greater risk of non-communicable diseases (NCDs) later in life, while present-day diet and other behavioural factors add further risk.

Foodgrain entitlement should not itself be equated with diets risks. The concern is that a predominantly cereal-based food basket, when combined with diets already high in carbohydrates and low in protein-rich foods, may reinforce dietary imbalance. The solution is not to reduce essential cereal entitlements, but to finance dietary diversification alongside.

The dietary evidence favours balance over simply increasing or substituting cereals. A 2025 ICMR-INDIAB study of 18,090 adults found carbohydrates supplied 62.3% of daily energy nationally and protein 12%. Adults with the highest carbohydrate intake had 30% higher odds of newly diagnosed type 2 diabetes than those with the lowest intake. Replacing refined cereals with whole-wheat or millet flour was not linked to lower risk when the carbohydrate share remained high. Modelling suggested greater benefit from replacing some carbohydrate calories with protein-rich foods.

These observational and modelled findings cannot establish that changing the Public Distribution System (PDS) basket will prevent diabetes, but caution against treating more grain or millets alone as a complete nutrition policy. The ICMR-NIN's 2024 guidelines recommend that cereals and millets provide at most 45% of energy, with greater contributions from pulses, beans, milk, nuts, vegetables, fruits and other protein sources.

The PDS cannot supply the entire healthy plate; it can best provide affordable, shelf-stable foods. States should be supported to offer pulses, local rice, wheat and millet choices and, where feasible, healthy edible oils. Local production and consumption should guide supply chains and effective Minimum Support Price (MSP) procurement for pulse, millet and oilseed growers. Sustained public investment should link food, nutrition and health systems through complementary programmes: the PDS, Anganwadi services for young children and pregnant or lactating women, and Pradhan Mantri Poshan Shakti Nirman (PM POSHAN) for

schoolchildren. These programmes could provide eggs, milk or suitable alternatives where locally acceptable and feasible. Dietary diversification must be separately budgeted, not financed by reducing cereal entitlements. The 2026-27 Union Budget allocates ₹2,27,629 crore to food subsidy, underscoring the need for transparent costing. Additions should be tested through phased State pilots assessing actual consumption, dietary diversity, anaemia and glycaemic risk, household spending, wastage, exclusion and supply feasibility. Pilot evidence can guide fiscally sustainable expansion and help assess potential longer-term health and fiscal benefits.

Linking food access with nutrition support

The NFSA's strength lies in guaranteeing reliable, dignified food access. India has built a formidable delivery network. By the end of 2025, 5.50 lakh of 5.51 lakh fair price shops used electronic point-of-sale devices, while One Nation One Ration Card covered nearly all NFSA beneficiaries. These systems support portability and monitoring but must include reliable offline alternatives, assisted or doorstep access for people with limited mobility, and a clear guarantee that authentication failure will not lead to denial of entitled foodgrains.

Fair price shops could support last-mile nutrition information and referral. Simple multilingual receipt, display, voice and mobile messages could promote dietary diversity, appropriate cereal portions, pulses, healthier oils, and less salt and added sugar. Where feasible, trained health workers could undertake periodic outreach through India's primary-health infrastructure, linking willing adults to government diabetes, hypertension and other NCD services. By June 2026, India had over 1.86 lakh Ayushman Arogya Mandirs, while government data recorded 41.3 crore diabetes screenings. Fair price shop dealers should facilitate information and referral, not diagnose or counsel. Participation must be voluntary, confidential and unrelated to ration eligibility or authentication.

The amendment should therefore be situated within a broader architecture for food and nutrition security, anchored in three safeguards. First, any per-person formula should preserve the existing 35 kg monthly entitlement for every AAY household through an explicit no-loss guarantee. Second, the adequacy of the 35 kg ceiling should be reviewed periodically for larger households and those with high dependency or nutritional needs, using consumption, nutritional and fiscal evidence. Third, dietary diversification must be separately financed and progressively implemented without reducing existing cereal entitlements.

India's next food security reform must protect people from hunger while also addressing the dietary drivers of diabetes and other NCDs. Its success should be judged not only by the tons of grain moved, but by whether vulnerable families can eat enough, eat healthier and obtain their entitlements with dignity.

A Timely Reset for the Food Security Act

- **The National Food Security Act of 2013 was enacted with the objective of providing food and nutritional security by ensuring access to adequate quantity of quality food at affordable prices for people to live a life with dignity.**
- **The Act covers up to 75% of the rural population and up to 50% of the urban population for receiving subsidized good grains under the Targeted Public Distribution System (TPDS), thus covering about two-thirds of the population.**
- **Under NFSA, the government is obliged to give subsidized food grains every month to the beneficiaries identified by each state government based on the economic status of households.**
- **Two categories of beneficiary households:**
 - **Antyodaya Anna Yojana (AAY): AAY households receive 35 kg of food grains monthly.**
 - **Priority Households (PHH): Each PHH member gets five kg of food grains monthly, including rice , wheat and coarse grain**

A Timely Reset for the Food Security Act

Other Coverage under NFSA

- **The free nutritious meal to pregnant women and lactating mothers and children up to 6 years of age under the Integrated Child Development Services (ICDS) scheme.**
- **The free nutritious meal to children in the 6-14 age group under the Mid-Day Meal Scheme.**
- **Maternity benefits of Rs. 6000 to Pregnant Women and Lactating Mothers.**

A Timely Reset for the Food Security Act

- **Proposed Amendments to NFSA**
 - **A new provision In Section 3 of the National Food Security Act, 2013, proposes that every person belonging to households covered under the Antyodaya Anna Yojana (AAY) shall be entitled to 7 kg of foodgrains every month to a maximum of 35 kg per household.**
 - **Earlier, it was 35 kg per AAY household, irrespective of the number of members in the household.**
 - **For AAY card holders, this allocation will be free of any charges.**
 - **Rational behind the proposal: Rationalise foodgrain allocation by addressing the imbalance whereby smaller households receive disproportionately higher per-capita entitlements than larger households.**

A Timely Reset for the Food Security Act

- **Malnutrition Trends (NFHS-6, 2023-24)**

- Stunting among children under five fell from 35.5% to 29.3%, but wasting and underweight rates barely changed.
- Only about 15% of children aged 6-23 months receive a minimally adequate diet.

- **The Diet-Diabetes Link**

- The ICMR-INDIAB study found carbohydrates supply 62.3% of daily energy nationally; high-carbohydrate diets are linked to 30% higher odds of newly diagnosed type-2 diabetes.
- ICMR-NIN's 2024 guidelines recommend that cereals and millets provide at most 45% of daily energy.

Way ahead : Dietary diversification must be separately financed and progressively implemented without reducing existing cereal entitlements.

The Work That Won Deepak Dhar the Dirac Medal

The work that won Deepak Dhar the Dirac Medal

The Dirac Medal is awarded to physicists who have made outstanding contributions to theoretical physics

Vasudevan Mukunth

The story so far:

Theoretical physicist Deepak Dhar has been awarded the Dirac Medal for his contributions to statistical physics – an honour he is sharing with three other physicists, Bernard Derrida, Marc Mézard, and Haim Sompolinsky. The Dirac Medal is awarded by the International Centre for Theoretical Physics (ICTP) in Trieste, Italy, on the birthday of the late British physicist Paul A.M. Dirac, for significant contributions to theoretical physics.

Dhar has spent most of his career in India. He is also the second Indian winner of the Dirac Medal; the first was string theorist Ashoke Sen in 2012.

In physics circles, Dhar is known for his work on a concept called Abelian sandpiles.

What is an Abelian sandpile?

Imagine dropping grains of sand, one at a time, on a flat surface. The pile slowly grows until it gets steep enough. Then, sometimes new grains on the pile will shift it, maybe cause a modest slide, sometimes even trigger an avalanche. And it is hard to predict in advance which grain will trigger which size of event.

In the late 1980s, physicists Per Bak, Chao Tang, and Kurt Wiesenfeld proposed that this behaviour could be a general principle underlying many phenomena in nature, from earthquakes to forest fires to the extinction of species. They called it self-organised criticality. The term refers to a system that sometimes organises itself into a delicately balanced critical state where even small nudges can produce effects of any size, from trivial to significant.

Dhar and his collaborators then developed the so-called Abelian sandpile

model as a mathematical version of the sandpile idea. 'Abelian' means no matter in which sequence grains topple and cascade, the pile always settles into the same final shape. This was Dhar's important insight – a property that lets physicists calculate exact answers and thus predict the final shape.

The Abelian sandpile model has numerous applications fundamentally because it captures a very general kind of behaviour: of systems that build up some kind of instability until they can release it in unpredictable ways. Scientists have used it to model earthquakes, the spread of forest fires, bursts of activity in neural networks in the brain, and fluctuations in financial markets.

Biographical sketch

Dhar was born in 1951 in Pratapgarh, Uttar Pradesh. He studied at Allahabad University, got his master's degree from

IIT-Kanpur, and then went to the California Institute of Technology (Caltech) for his PhD, which he completed in 1978. While at Caltech, he was a teaching assistant to Richard Feynman.

After finishing his doctorate, Dhar returned to India rather than stay in the U.S., and spent most of his career at the Tata Institute of Fundamental Research, Mumbai, before moving to the Indian Institute of Science Education and Research (IISER), Pune, in 2016. Since 2024, he has been the INSA Distinguished Professor at the International Centre for Theoretical Sciences, Bengaluru.

The Abelian sandpile is Dhar's most well-known work but not his only one. Across nearly five decades, Dhar has made important contributions to statistical mechanics – the branch of physics concerned with how the collective behaviour of vast numbers of particles, like grains or neurons, gives rise to large-scale patterns.

He won the Shanti Swarup Bhatnagar Prize in 1991, the TWAS Prize from the World Academy of Sciences in 2002, the Padma Bhushan in 2023, and the Boltzmann Medal in 2022, together with John Hopfield, who shared the physics Nobel Prize 2014 for laying the foundations of neural networks. The Boltzmann Medal, awarded once every three years, is often considered the single most prestigious prize in statistical

physics.

What is the Dirac Medal?

The ICTP Dirac Medal was established in 1985 to honor Paul Dirac, the British physicist whose equations helped lay the foundations of quantum mechanics. It is announced each year on Dirac's birthday and awarded to physicists who have made outstanding contributions to theoretical physics but who have not already received the very top honours.

In fact, the medal is explicitly not given to Nobel laureates, Fields Medalists, or Wolf Prize laureates. Past recipients include Edward Witten, Stephen Hawking, and Leonard Susskind. Notably, several past Dirac Medalists have gone on to win a Nobel Prize.

All four of this year's winners have been feted for pioneering work that applied the tools of statistical mechanics in distinct fields like non-equilibrium systems, optimisation problems, theoretical neuroscience, and artificial intelligence.

The ICTP itself was founded in 1964 by the Pakistani physicist and Nobel laureate Abdus Salam, together with the Italian government and the International Atomic Energy Agency. Salam had envisioned it as a place where physicists from the Global South could work together and exchange ideas with colleagues from around the world.

The Work That Won Deepak Dhar the Dirac Medal

- **What is the Dirac Medal**

- Established in 1985 by the International Centre for Theoretical Physics (ICTP), Trieste, to honour physicist Paul A.M. Dirac.
- Awarded annually on Dirac's birthday to physicists with outstanding contributions to theoretical physics who have not already received the very top honours, it is explicitly not given to Nobel laureates, Fields Medallists, or Wolf Prize laureates.

- **About ICTP**

- Founded in 1964 by Pakistani physicist and Nobel laureate Abdus Salam, envisioned as a hub for physicists from the Global South to collaborate with peers worldwide.

Kazakhstan, Desperate for Rain, Turns to Cloud-Seeding



An engineer mounts hypersonic laser-attracting salt flares on a wing of a cloud seeding plane in southern Kazakhstan. AP

Kazakhstan, desperate for rain, turns to cloud-seeding

Agence France Presse

There were only arid droplets in sight when a cloud plane took off from an airport in northern Kazakhstan, embarking on a delicate mission: bringing rain for the parched earth below.

In tackle the severe drought and water shortages that plague the country, Kazakhstan has resorted to cloud seeding, a weather modification method previously unseen in Central Asia.

"The aircraft is specifically equipped with a cloud seeding flare system," Abnol Alabert, a pilot with the United Arab Emirates' meteorological agency, which conducts cloud seeding at home, well is collaborating with Kazakhstan on the project, explained.

Cartridge fixed to the aircraft's wings contain a substance that, when dispersed, sticks to cloud droplets, making them heavy enough to fall as rain.

Kazakh authorities are betting on cloud seeding to reduce the impact of drought and water shortages in its southern Turkestan area, the country's leading cotton-growing region.

Areas in the region with some 5,000 square kilometers of arable land are being targeted, with estimated economic benefits of around \$15 million a year through better harvests.

Kazakh authorities are betting on cloud seeding to reduce the impact of drought and water shortages in its southern Turkestan area, the country's leading cotton-growing region

Delivering droplets onto the parched crops require intricate cloud penetration to identify suitable clouds.

"The plane must 'get as close as possible, detect the updrafts, and begin seeding,'" Mr. Alabert said on board the flight.

At best, the technique can increase precipitation by 15% to 20%, according to various scientific studies.

But those involved also highlight the significant risk of failure due to numerous factors, not least because not just any cloud can be seeded.

"Salt particles – specifically sodium chloride and potassium chloride – are most often used" as cloud seeding agents, said Abnol Yurandzhennel, one of the Russian officials in charge of sharing his country's experience.

The techniques vary either seeding rockets are burned to release a plume of particles, or the saline solution is sprayed directly into the cloud for greater precision.

"This 'weighs down the droplets,'" which "grow from a few micrometres to several millimetres, up to two or three. They fall, and it rains," explained Mr. Yurandzhennel.

He added that the technique "does not create new clouds".

The Kazakh meteorological service said the impact of the project is "local and short-lived" and that it "does not affect climate processes or alter the movement of air masses".

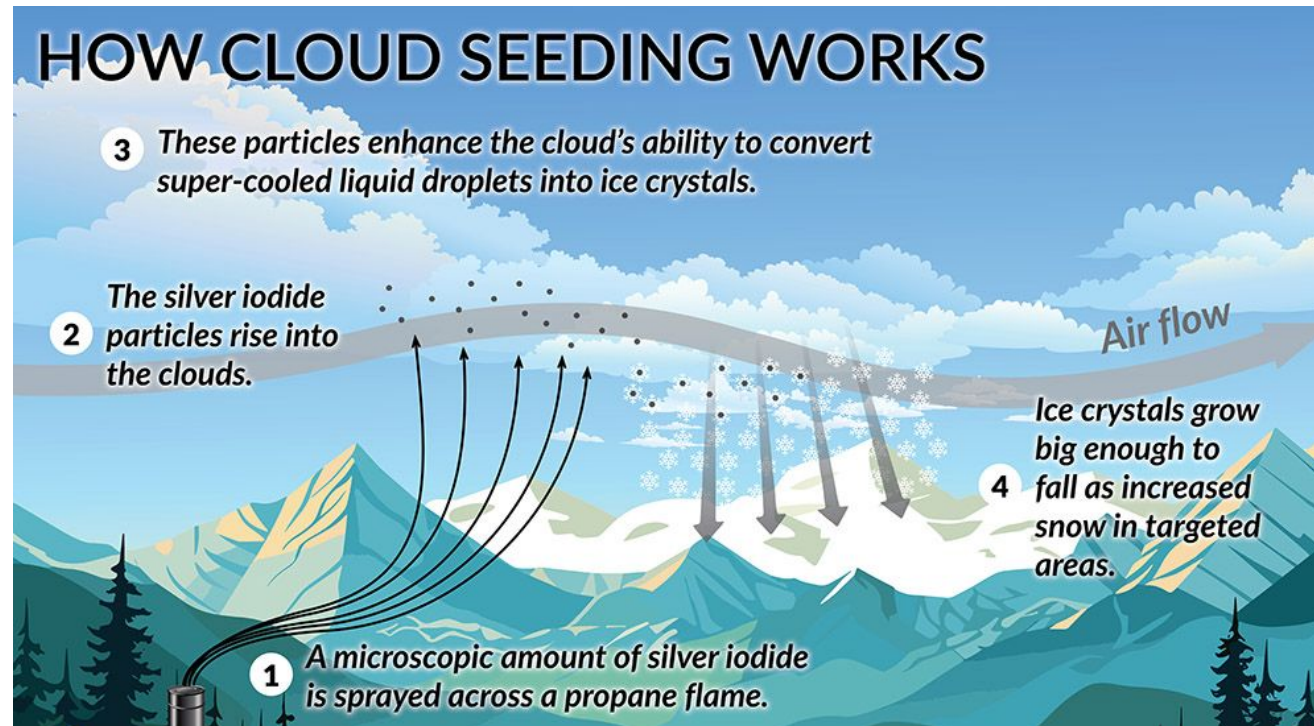
The effects of boosting rainfall through cloud seeding remain uncertain, its consequences difficult to measure, and its large-scale application costly.

The World Meteorological Organization has called for more scientific research, pointing to a "rise in demand for these activities ... due to the frequency of droughts".

The UN organisation has also ruled out any danger to human health or the environment and acknowledged potential economic benefits if the technique proved successful. The method, already used in around 50 countries to combat drought, hail and air pollution, is new for Central Asia – a region warming twice as fast as the global average.

Kazakhstan, Desperate for Rain, Turns to Cloud-Seeding

- Cloud seeding is a weather modification technique aimed at enhancing precipitation (rain or snow) from existing clouds.
- Mechanism: It involves dispersing substances such as silver iodide, potassium iodide, sodium chloride, or dry ice (solid CO₂) into clouds using aircraft.



Kazakhstan, Desperate for Rain, Turns to Cloud-Seeding

- Prelims

Q. Artificial way of causing rainfall to reduce air pollution makes use of

- A. silver iodide and potassium iodide
- B. silver nitrate and potassium iodide
- C. silver iodide and potassium nitrate
- D. silver nitrate and potassium chloride

The Next Step to the Moon — ISRO and NASA's Moon Base

The next step to the Moon

AMID THE volatility of India-US relations, the one aspect of the partnership that has continued to make steady progress is space cooperation. The two space agencies, ISRO and NASA, have never worked so closely together. Now, NASA has asked ISRO to join its Moon Base programme, an ambitious project to establish a permanent research station on the Moon, where astronauts can live, work and carry out experiments for extended periods of time. Humans haven't been to the Moon since 1972, and efforts have now begun to stage a return, this time to prepare the ground for longer stays. Unsurprisingly, the effort is being spearheaded by the US, through its Artemis programme, which aims to land humans on the Moon before 2028. The Moon Base programme is the logical follow-up, NASA realises this goal could be achieved much faster, and more efficiently, if other partner countries and private companies join the effort. ISRO, which has its own plans to land human beings on the Moon by 2040, will get valuable hands-on experience in executing complex crewed missions, and access to technologies. India has already signed up to the Artemis Accords, a US-led coalition of space-faring countries. India and the US have also agreed to develop a "strategic framework for human spaceflight cooperation".

The exploration of the Moon is no longer just a scientific enterprise. Over the next few decades, the Moon could become strategically and economically very important, as countries begin to extract lunar resources. But there are potential pitfalls as well. The Artemis Accords are increasingly being seen as a US-led alliance. Two major space powers, China and Russia, are not part of it. The Accords sidestep, and seek to replace, the 1979 Moon Agreement, which seeks to develop a multilateral governance framework for use of lunar resources.

India has been wary of joining such international structures. Having signed on to the Artemis Accords, however, it has made a choice. Accepting the offer to join the Moon Base programme could be the next logical step forward. But it is extremely important that ISRO does not get locked into NASA's technology ecosystem. That would once again make ISRO vulnerable to technology denial. The cooperation on the Moon Base programme should help ISRO to achieve its own goals faster and more efficiently, and not abandon or delay them in the service of someone else's goals.

The Next Step to the Moon — ISRO and NASA Moon Base

- **ARTEMIS Mission**

- **Artemis Mission is a lunar mission to establish the first long-term presence on the Moon.**
- **The mission intends to re-establish human presence on the Moon for the first time since the Apollo 17 mission in 1972.**
- **NASA partnered with three space agencies: the European Space Agency (ESA), the Japan Aerospace Exploration Agency (JAXA) and Canadian Space Agency (CSA).**
- **Artemis mission is divided into four parts: Artemis I, Artemis II, Artemis III, and Artemis IV.**

Thank You